Trainings

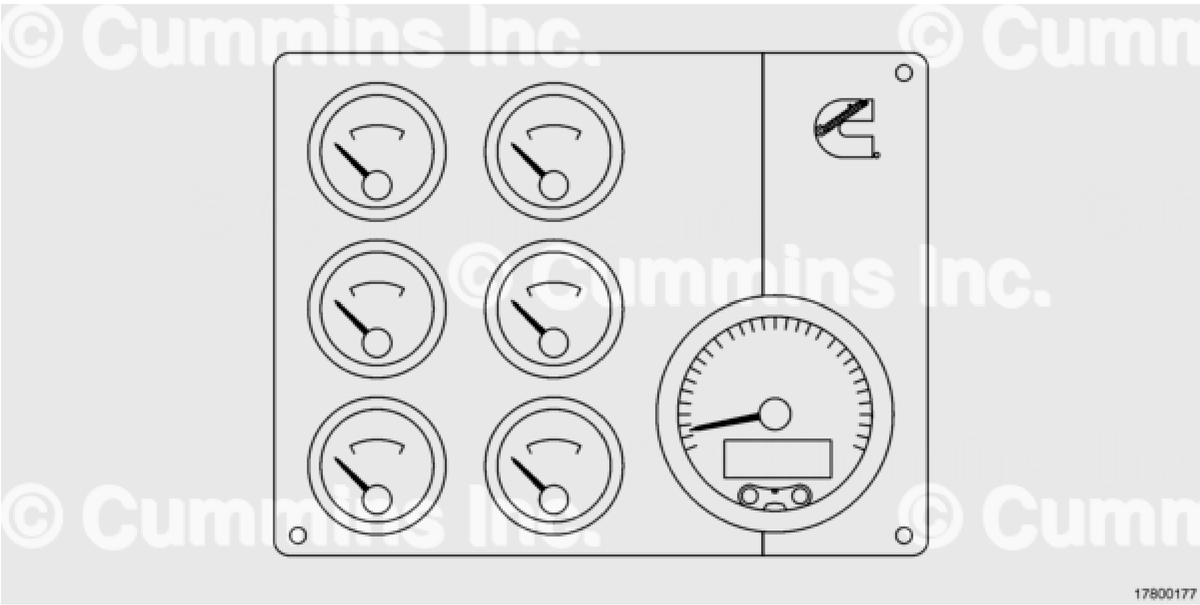
General Information

The gauges have been developed for applications where the SAE J1939 datalink communication bus is available.

The gauge components of the system are based on stepper motor technology, integrated with a microcontroller that provides the necessary electrical control. The primary gauge size is 85 mm [3.23 in], and the secondary gauge size is 52 mm [2.05 in] in diameter. The instrument enclosure can have the connector molded in as an integral part. The gauge sets have one primary gauge and up to six secondary gauges.

The primary gauge includes all communications, power supply, memory, display, and electronics to control the system.

The secondary gauges, including the warning bank modules, will receive power and communications from the primary device.



Lighting of the dial face is provided by light emitting diodes. The dial is a through illumination. The backlighting of the gauge is controlled by the gauge's microcontroller via a pulse width modulated output. Standard backlight color is red.

The pointer length for the primary gauge is 44 mm [1.73 in] and for the secondary gauge is 19 mm [0.75 in]. The pointer is illuminated when the gauge backlight is on. Standard pointer color is orange. The primary gauge pointer can perform a 230 degree sweep and the secondary gauge pointer can perform a 270 degree sweep. The pointer hub is black.

In each gauge a microprocessor controlled stepper motor will position the pointer to within an angular degree of the position determined by the data received from the SAE J1939 datalink over the entire operating temperature and voltage range.

The message center located on the primary gauge, is a full graphic display. The specifications are described in the table below:

Gauge Display Specifications	
Description	**Specification**
Viewable area	51.6 mm [2.15 in] x 19.05 mm [0.75 in]
Pixel array	122 x 32 pixels
Polarizer mode	Transflective positive
Backlighting	Custom
Operating temperature	-20°C [-4°F] to 70°C [158°F]
Storage temperature	-40°C [-40°F] to 85°C [185°F]
Backlight brightness	Pulse width modulated controlled by gauge's microprocessor

The user can interact with the gauge set through the mode and trip buttons on the front of the primary gauge.

There is an audible alarm in back of the primary gauge. The tones are assigned to warning conditions received from the SAE J1939 datalink. Each secondary gauge has a lamp that illuminates when an alarm occurs.

There are two communications channels for data transfer between the vessel and primary gauge. One channel for SAEJ1708/J1587 datalink and second channel for SAE J1939 datalink. A selectable termination resistor is available for the SAE J1939 datalink.

A local interconnect network communication channel is used for communication between the primary gauge and the secondary gauges.

The main power input for the primary gauge is 9 to 32 VDC for normal operation. Voltage less than 9 VDC will cause the microcontroller to shut down because of low voltage.

The main power input for the secondary gauge is 7.5 VDC and supplied by the primary gauge. This supply is protected against EMC, over voltage, reverse polarity, and short circuit.

The following table are for connector assignments:

Connector to Secondary Devices		
Terminal	**Designation**	**Description**
1	(+) Power	7.5 VDC power output
2	GND	Device ground

Connector to Secondary Devices		
Terminal	**Designation**	**Description**
3	LIN	Local interconnect network for secondary functions
4	Open	**Not** used

Power and Communication Connector		
Terminal	**Designation**	**Description**
1	(+) Battery	Unswitched positive battery input
2	Ground	Chassis ground
3	(+) CAN	SAE J1939 datalink supply
4	(-) CAN	SAE J1939 datalink return
5	(+) J1708	SAE J1708 datalink supply
6	(-) J1708	SAE J1708 datalink return
7	Reserved	Reserved for ISO 9141 diagnostic communication port.
8	CAN Res	SAE J1939 datalink termination resistor.

Primary Gauge Input and Output Connector			
Terminal	**Designation**	**Type**	**Description**
1	AN3	Voltage or resistance	Analog input
2	AN1	Voltage or resistance	Analog input
3	AN2	Voltage or resistance	Analog input
4	AN0	Voltage or resistance	Analog input with wake-up capability
5	AN4	Voltage or resistance	Analog input
6	AN5	Voltage or resistance	Analog input
7	AN6	Voltage or resistance	Analog input
8	AN7	Voltage or resistance	Analog input
9	BIN_OUT1	Low side 1 amp	Binary output
10	BIN_OUT2	Low side 1 amp	Binary output
11	IGNITION	High side	Binary input for ignition switch

Primary Gauge Input and Output Connector			
Terminal	**Designation**	**Type**	**Description**
12	BIN0	Pull-up or pull-down	Binary input with wake-up capability
13	BIN1	Pull-up or pull-down	Binary input with wake-up capability
14	BIN2	Pull-up or pull-down	Binary input
15	BIN3	Pull-up or pull-down	Binary input
16	BIN4	Pull-up or pull-down	Binary input

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